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Paper Code : PEC-CS701E/PEC-IT701D/PECICB702A Machine learning
UPID : 007554

CS/B.TECH(N)/ODD/SEM-7/7554/2025-2026/1019

Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

[1 x 10 = 10]

- (I) What is the main objective of an agent in reinforcement learning?
- (II) In reinforcement learning, what term is used for the signal that indicates the quality of an action taken?
- (III) Give an practical example of reinforcement learning.
- (IV) Logistic regression is generally used for which type of classification?
- (V) What does PCA stands for?
- (VI) Which ensemble method involves training multiple models in such a way that each model corrects the errors of the previous model?
- (VII) In deep learning, what is the role of dropout regularization?
- (VIII) How does distributed learning allow for scalability in training large datasets?
- (IX) What is continual learning in ML, and why is it important?
- (X) What is the use of cross validation.
- (XI) How does L1 regularization encourage sparsity in model coefficients?
- (XII) In ensemble methods, what is the main advantage of boosting over bagging?

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 x 3 = 15]

2. Describe the K-means clustering algorithm. ✓ [5]
3. What is the benefit of Random forest classifier over individual decision trees. [5]
4. What is the primary goal of sparse modeling and how does it differ from traditional modeling approaches? ✓ [5]
5. Define a Bayesian network and explain how it can be used to represent probabilistic relationships between a set of variables. [5]
6. Explain the concept of overfitting in machine learning. How can techniques like cross-validation help in mitigating it? ✓ [5]

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 x 3 = 45]

7. (a) Explain the importance of feature selection in supervised machine learning. What techniques can be used for feature selection and what factors should be considered when choosing features? [6]
(b) Compare and contrast classification and regression tasks in supervised machine learning. Provide examples of each. [4]
(c) Describe linear regression as a supervised learning algorithm. How is the best-fit line determined and how are predictions made with it? [5]
8. (a) Define dimensionality reduction and explain its significance in data processing. [5]
(b) Discuss problems associated with high-dimensional data, highlighting the concept of the "curse of dimensionality". [5]
(c) Explain the relationship between the variance of the data and the principal components derived using PCA. [5]
Discuss a limitation of PCA when it comes to capturing non-linear relationships in the data.
9. (a) Explain Generative Mixture model. [5]
(b) With a proper diagram explain the steps of a generative mixture model. [5]
(c) What are parametric and non-parametric model. [5]

10. (a) Matrix factorization plays a significant role in collaborative filtering-based recommendation systems. Explain how matrix factorization works in the context of recommendation systems. [7]
(b) Discuss the challenges of matrix completion when dealing with sparse user-item interaction data. [5]
(c) Describe one matrix factorization technique used for recommendation. [3]
11. (a) Define what bagging is and explain its core principle. Describe how it reduces overfitting and variance in machine learning models. [5]
(b) List two benefits and one potential drawback of using bagging in the context of decision trees. [5]
(c) Discuss the concept of bootstrapping in bagging and explain how it contributes to model diversity. [5]

END OF PAPER